Personal details and date of CV

- **Surname:** Zakary
- **First name:** Ouail
- **ResearcherID:** OSI-7733-2025
- **ORCID:** 0000-0002-7793-3306

- **Date of the CV:** 28.6.2026

Degrees

- **08.12.2023:** Ph.D. in Physics, Dept. of Physics, *Le Mans Université* (*MU*), Le Mans, France.
- **30.06.2020:** M.Sc. in Physics, Dept. of Physics, *MU*, Le Mans, France.
- **30.06.2020:** M.Sc. in Physics, Dept. of Physics, *Hassan II University of Casablanca* (*H2UC*), Casablanca, Morocco.
- **30.06.2018:** B.Sc. in Physics, Dept. of Physics, *H2UC*, Casablanca, Morocco.

Language Skills (5)

- **English** and **French** (Proficient, C2)
- **Arabic**, **Moroccan Arabic/Darija**, and **Amazigh/Tamazight** (Native)

Current employment

- **05.02.2024–04.02.2027:** Postdoctoral Researcher, *NMR Research Unit* (*NMRU*), *University of Oulu* (*UO*), Finland (Website: https://cc.oulu.fi/ nmrwww/index.html). Developing machine learning methodologies for NMR spectroscopic parameters, interatomic potentials, molecular and solid-state simulations; co-supervising Ph.D., M.Sc., and B.Sc. students and teaching computational physics and chemistry courses.
- **Stage II** of academic research career on the four-stage (I–IV) research career model of Finland.

Previous work experience

- **01.10.2020–08.12.2023:** Doctoral Researcher (PhD Student), Department of Physics, *Institut des Molécules et Matériaux du Mans* (*IMMM–UMR 6283 CNRS*), *MU*, Le Mans, France. **Research focus**: Structural modeling of disorder in transition metal oxyfluorides (thesis).
- **04.04.2022–08.04.2022:** Visiting PhD Student working on solid-state NMR with Dr. Vincent Sarou-Kanian (ResearcherID) at the *CEMHTI* lab, Orléans, France.
- **01.05.2020–31.08.2020:** Internship at *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France.
- **01.06.2019–31.08.2019:** Internship at the *Laboratoire de Physique de la Matière Condensée* (LPMC), *H2UC*, Casablanca, Morocco.

Research funding and grants

- **2024:** 3300 € from *Otto A. Malm Foundation*, travel grant for the project "*Realistic Modeling of Large-Scale Porous Materials using Machine Learning-Assisted Molecular Dynamics*".
- **2020–2023:** Excellence Grant, 6000 € from the Ministry of Higher Education and Research of Morocco, awarded for academic merit during doctoral studies at *MU*, Le Mans, France.

Research output

I. Publications:

- **Total:** 5 published scientific articles, 1 article submitted for publication, and 11 in preparation.
- **10 most important publications** (corresponding authorship marked with *):

1. **Zakary, O.***; Yin, W.; Aryal, N.* Local Symmetry Breaking and Two-Stage Phase Transition in

RuP Uncovered by a Fine-Tuned Atomistic Foundation Model. *ChemRxiv* **2026**, 0330, *Submitted for publication*.
[DOI: 10.26434/chemrxiv.15001387/v1] [Supporting Data] [Supporting Code]
2. Laurila, O.; Jacklin, T.; **Zakary, O.***; Lantto, P.* Machine Learning-Accelerated Path Integral Molecular Dynamics and ^{13}C NMR Simulations Unlock New Insights Into Quantum Effects in C_{60} Fullerene. *J. Phys. Chem. A* **2026**, 130, 2169–2181.
[DOI: 10.1021/acs.jpca.6c00238] [Supporting Data] [Supporting Code]
3. **Zakary, O.***; Lantto, P. Equivariant Neural Networks Reveal How Host–Guest Interactions Shape ^{129}Xe NMR in Porous Liquids. *J. Phys. Chem. Lett.* **2025**, 16, 12095–12103. (Selected as *Journal Supplementary Cover*)
[DOI: 10.1021/acs.jpclett.5c02846] [Supporting Data] [Supporting Code] [Journal Cover]
4. **Zakary, O.***; Body, M.; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Revealed Preferential Short-Range Anion Ordering in Disordered RbM_2O_5F (M= Nb, Ta) Pyrochlore-Type Oxyfluorides. *Inorg. Chem.* **2025**, 64, 5764–5777.
[DOI: 10.1021/acs.inorgchem.5c00615] [Supporting Data] [Supporting Code]
5. **Zakary, O.***; Body, M.; Sarou-Kanian, V.; Arnaud, B.; Corbel, G.*; Legein, C. Different Magnitudes of Second-Order Jahn-Teller Effect in Isostructural $NaMO_2F_2$ (M= Nb, Ta) Oxyfluorides. *J. Alloys Compd.* **2025**, 1010, 177457.
[DOI: 10.1016/j.jallcom.2024.177457] [Supporting Data] [Supporting Code]
6. **Zakary, O.***; Body, M.*; Charpentier, T.; Sarou-Kanian, V.; Legein, C. Structural Modeling of O/F Correlated Disorder in $TaOF_3$ and $NbOF_{3-x}(OH)_x$ by Coupling Solid-State NMR and DFT Calculations. *Inorg. Chem.* **2023**, 62, 16627–16640.
[DOI: 10.1021/acs.inorgchem.3c02844] [Supporting Data]
7. **Zakary, O.***; Jacklin, T.; Lantto, P. Transport Properties in Carbon Nanotubes Unlocked by Machine Learning-Accelerated Molecular Dynamics and Noble Gas NMR Simulations. 2026, *In preparation*.
8. **Zakary, O.***; Mailhiot, S. E.; Egleston, B. D.; Selent, A.; Kearsey, R. J.; Mareš, J.; Cooper, A. I.; Greenaway, R. L.; Telkki, V.-V.*; Lantto, P.* Unifying the Description of Type-II Porous Liquids Through Machine Learning-Accelerated Molecular Dynamics and ^{129}Xe NMR: Simulations Informing Experiments. 2026, *In preparation*.
9. **Zakary, O.*** A Machine Learning Workflow for Solving Disorder in Heteroanionic Materials. 2027, *In preparation*.
10. **Zakary, O.*** A Foundation Model for NMR Parameters in Materials Chemistry. 2027, *In preparation*.

II. Software and Code Development:

- **Total:** 30 *Github* repositories (25 public and 8 private, more details can be found here).
- **10 most important repositories**:

1. **Lammps-Kokkos-Mace_Mahti-CSC** – This repository provides a recipe for the installation of LAMMPS with Kokkos GPU acceleration and MACE model for ML interatomic potential support on the Mahti CSC supercomputer.
[Repository: github.com/ozakary/Lammps-Kokkos-Mace_Mahti-CSC]
2. **Lammps-Kokkos-Mace_Puhti-CSC** – This repository provides a recipe for the installation of LAMMPS with Kokkos GPU acceleration and MACE model for ML interatomic potential support on the Puhti CSC supercomputer.
[Repository: github.com/ozakary/Lammps-Kokkos-Mace_Puhti-CSC]
3. **Lammps-Kokkos-MACE-single-GPU_LUMI** – This repository documents how to build and run LAMMPS with the native *PKG_ML-MACE* interface for MACE machine learning force field simulations on the LUMI supercomputer using a single AMD MI250X GPU.
[Repository: github.com/ozakary/Lammps-Kokkos-MACE-single-GPU_LUMI]

4. **Lammps-Kokkos-Symmetrix-multi-GPU_LUMI** – This repository documents how to build and run LAMMPS with the *pair_symmetrix* interface for MACE-based machine learning interatomic potentials on the LUMI supercomputer using AMD MI250X GPUs.
[Repository: github.com/ozakary/Lammps-Kokkos-Symmetrix-multi-GPU_LUMI]
5. **MACE_hyperparams** – This repository contains the methodology and tools for optimizing the hyperparameters of the Equivariant Message Passing Neural Network model MACE.
[Repository: github.com/ozakary/mace_hyperparams]
6. **NMR-MatTen** – Python-based workflow for generating an NMR-ML model using the *E(3)-Equivariant Graph Neural Networks* architecture *MatTen*.
[Repository: github.com/ozakary/NMR-MatTen]
7. **NMR-VASP** – VASP NMR parameter calculator that extracts and processes NMR data from OUTCAR files for material science applications.
[Repository: github.com/ozakary/NMR-VASP]
8. **TrajSlicer** – Python tool for converting LAMMPS trajectories to XYZ format.
[Repository: github.com/ozakary/TrajSlicer]
9. **xyz-proton-tracker** – Python tool for analyzing molecular dynamics trajectories in XYZ format, specifically designed to identify and visualize different protonation states of oxygen atoms (OH^-, H_2O, H_3O^+) through color coding for OVITO visualization.
[Repository: github.com/ozakary/xyz-proton-tracker]
10. **RDFComp** – A command-line tool for computing radial distribution functions (RDFs) from molecular dynamics trajectories in XYZ format, powered by OVITO.
[Repository: https://github.com/ozakary/RDFComp]

III. Artistic and Visual Outputs:

1. **Journal Supplementary Cover Design** – Designed supplementary cover for *The Journal of Physical Chemistry Letters*, Volume 16, Issue 46 (2025).
[Published Cover]

Research Supervision and Leadership Experience

I. Supervision of Students:

i. Ph.D. (1):

- Ossi Laurila (2025–present), *NMRU*, *UO*, Oulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Machine learning methods for investigating nuclear quantum effects.

ii. M.Sc. (6):

- Matias Hintsanen (2026), *NMRU*, *UO*, Oulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Insights into the Density Maximum of Water from Machine Learned ^{129}Xe NMR and Molecular Dynamics.
- Ossi Laurila (2024), *NMRU*, *UO*, Olulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Machine learning methods for investigating nuclear quantum effects.
- Rachid Belrhiti-Nejjar (2022), *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* Solid-state NMR of paramagnetic copper and nickel phenanthroline complexes.
- Samira Tepondjou (2022), *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* Characterization of hybrid perovskites by solid-state ^{207}Pb NMR.
- Odéric Claverie (2023), *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France. Co-supervised with Prof. Jens Dittmer. *Topic:* NMR correlation experiments for the study of whole diatom cells enriched with ^{13}C and ^{15}N.
- Rim Hussein (2023), *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France. Co-supervised with Prof.

Jens Dittmer. *Topic:* NMR of microscopic-sized single crystals.

iii. B.Sc. (1):

- Matias Hintsanen (2025), *NMRU*, *UO*, Oulu, Finland. Co-supervised with Dr. Perttu Lantto. *Topic:* Insights into the Density Maximum of Water from Machine Learned ^{129}Xe NMR and Molecular Dynamics.

***II. Supervision Responsibilities*:** Research methodology development, progress monitoring and troubleshooting, manuscript preparation guidance, and training in computational physics and chemistry.

Teaching merits

***I. Pedagogical Training*:** *Pédagogies de l'enseignement supérieur* (Pedagogies of Higher Education), *MU*, Le Mans, France (2020).

***II. Teaching experience*:** 155 hours across five courses at B.Sc., M.Sc., and Ph.D. levels.

- **Computational Physics and Chemistry** (M.Sc./Ph.D. level, 16 h), Faculty of Science, *UO*, Oulu, Finland (2024–2025).
- **Simulation of Phenomena in Physical Sciences using Python** (B.Sc. level, 24 h), Faculté des Sciences et Techniques, *MU*, Le Mans, France (2022–2023).
- **Electromagnetism: Measurements and Instrumentation** (B.Sc. level, 66 h), **Wave and Physical Optics** (B.Sc. level, 43 h), and **Learning and Evaluation Situation** (B.Sc. level, 6 h), IUT du Mans, *MU*, Le Mans, France (2020–2022).

***III. Teaching material development*:** Developed open-source GUI – *water-pimd-simulations*

Other key academic merits

- **2025**: Assisted in organizing the EUROMAR 2025 conference, the largest magnetic resonance conference in Europe.
- **2024–present**: Member of **follow-up groups** for Ph.D. students Anand Chekkottu Parambil (defended February 2026) and Piotr Sobiecki (expected defense 2028), *UO*, Oulu, Finland.
- **2025–present**: Member of the *AI Group for Research*, *UO*, Oulu, Finland.
- **2024–present**: Member of the *Magnetic Resonance in Porous Media* (MRPM) community.
- **2024–present**: Member of the *Groupment AMPERE* association.
- **2024–present**: Member of the *Finnish Union of University Researchers and Teachers*.
- **2024–present**: Responsible for NMRU website (updates, maintenance, *etc.*).
- **2020–2023**: Member of *L'Association des Doctorants de l'Université du Mans* (*ADoUM*), *MU*, Le Mans, France. Promoting networking, interdisciplinary interaction, and scientific outreach among PhD students.

Scientific and societal impact

***I. Science communication and public engagement*:**

- **2026**: Invited speaker at *Science Day* (organized by *Sigma-kilta*); presented an accessible talk, "*How Atoms Find Their Way: A Journey Through Energy Landscapes*," to a broad student audience to promote interest in physics and mathematics.
- **2023**: Organized and led laboratory visits and research demonstrations for high school students at *IMMM–UMR 6283 CNRS*, *MU*, Le Mans, France.
- **2020–2023**: Active participation in monthly science outreach events through *ADoUM*, including laboratory tours and *Café-Thèse* sessions for research dissemination to diverse audiences.